LEARNING SESSION S10

Reverse Engineering for Non-Tech Learners

Learn fast by taking finished work apart: dashboards, Excel models, SQL, notebooks, ML models, metrics and AI prompts

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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Learning session · a skill-building method used by analysts to understand inherited work and learn from good examples

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#s10**

The method

Reverse engineering means starting from a finished result and working backwards to understand how it was built. For a non-technical learner it is the fastest way to learn: instead of a blank page, you study a working example, rebuild a small version, and compare.



Six steps that work for any artefact.

Step	Questions to ask
1 Observe	What question does this answer? Who uses it? Which numbers matter most?
2 Decompose	What are the parts (pages, sheets, CTEs, cells, features)? Which part feeds which?
3 Trace	For one number: which data, which filters, which formula produced it?
4 Rebuild	Recreate one part on a small sample yourself
5 Compare	Do your numbers match? If not, what is different (filter, join, definition)?
6 Document & improve	Write the logic in plain English; note risks and improvements

> Only reverse-engineer work you are allowed to access: your company's files, public examples (Tableau Public, Kaggle notebooks, GitHub), or your own course material. Respect licences, terms of service and confidentiality.

Excel workbooks

- **Show Formulas** (`Ctrl` + `` ` ``) to see all logic at once.
- **Trace Precedents / Dependents** (Formulas tab) draws arrows from inputs to outputs; **Evaluate Formula** steps through a formula one calculation at a time.

- **Name Manager** lists named ranges and LAMBDA functions; **Go To Special → Formulas / Constants** highlights hard-coded numbers hiding among formulas.
- Data → Queries & Connections shows Power Query steps; Data → Relationships / Power Pivot shows the data model.
- **Inquire** add-in (Workbook Analysis, Compare Files) or Copilot's "Explain this formula".

Exercise

Take a finished dashboard workbook (e.g. a free Excel dashboard template). Pick the headline KPI, trace it back to raw data, write the calculation in one sentence, then rebuild it on a new sheet with your own formula and check it matches.

SQL queries

1. Format the query (SSMS: Edit → Advanced; or an online formatter) so each clause is on its own line.
2. Read in **execution order**: FROM/JOIN → WHERE → GROUP BY → HAVING → SELECT → ORDER BY.
3. For nested queries, read the **innermost** subquery first; for CTEs read top to bottom and run each CTE alone: `WITH a AS (...) SELECT * FROM a;`
4. Check row counts after each join to spot duplication.
5. Replace the final SELECT with `SELECT TOP 20 *` at each stage to see intermediate data.
6. Look at the execution plan (`Ctrl` + `M`) to understand how tables are combined.
7. Ask an AI assistant: "Explain this query line by line for a beginner and describe the business question it answers."

Dashboards

Power BI

- Model view: tables, relationships, direction.
- Select a visual → see fields in the Build pane; check visual/page/report filters.
- Right-click a measure → see its DAX; use the DAX query view to evaluate it alone.
- Performance Analyzer → Copy query shows the exact DAX each visual runs.
- Open a .pbip project to read the model as TMDL text.

Tableau

- On Tableau Public, many authors allow "Download" of the .twbx: open each sheet and look at shelves, the Marks card and filters.
- Right-click calculated fields → Edit to read formulas; check Compute Using on table calcs.
- Dashboard → Actions shows interactivity.

Exercise

Pick a Tableau Public "Viz of the Day". Identify its chart types, the calculated fields it must need, and rebuild one chart with a similar public dataset.

Notebooks & ML models

Python notebooks (Kaggle, GitHub)

- Run cell by cell; after each, print `df.shape`, `df.head()` to see what changed.
- Write a one-line comment above each cell in your own words.
- Change one parameter at a time and observe the effect.

ML models

What does it predict?	Find the target column and the business decision it supports
What drives it?	Feature importance, coefficients, SHAP summary plots
How does it behave?	Score a few hand-made examples ("what if income doubles?") – sensitivity analysis
How good is it?	Recompute the confusion matrix on a hold-out sample
Is it fair?	Compare error rates across groups

Metrics, apps & AI

Reverse-engineer a business metric

Build a **metric tree**: Revenue = Orders × AOV; Orders = Visitors × Conversion; AOV = Items per order × Price per item. When a number moves, walk down the tree to find which branch moved.

Reverse-engineer a product (case-study thinking)

Look at a feature (e.g. "Frequently bought together") and ask: what data does it need (order history), what algorithm (association rules), what metric would prove it works (attach rate, AOV)? This is how product and data interview cases are answered.

Reverse prompting (AI)

Give an AI an output you like (a report, an email, a chart description) and ask: "Write the prompt that would produce this." Save the resulting prompt as a template. Covered in **Generative AI → PT4 Reverse prompting**.

Public web data (responsibly)

Browser developer tools (F12 → Network) show the data a public page loads (often JSON). Use official APIs or downloads where provided, follow the site's terms, and never collect personal data.

Resources

- ▶ YouTube: Excel trace precedents evaluate formula tutorial · ▶ YouTube: how to read complex SQL queries
- ▶ YouTube: Power BI performance analyzer tutorial · ▶ YouTube: Tableau Public download workbook learn
- ▶ YouTube: SHAP values explained simply
- 📘 Tableau Public Viz of the Day · 📘 Kaggle Code (public notebooks)